

12.03	Analysing data				
12.03a	Summary statistics	Calculate the mean, mode, median and range for ungrouped data. Find the modal class, and calculate estimates of the range, mean and median for grouped data, and understand why they are estimates. Describe a population using statistics. Make simple comparisons. Compare data sets using 'like for like' summary values. Understand the advantages and disadvantages of summary values.		Calculate estimates of mean, median, mode, range, quartiles and interquartile range from graphical representation of grouped data. Draw and interpret box plots. Use the median and interquartile range to compare distributions.	S4, S5
12.03b	Misrepresenting data	Recognise graphical misrepresentation through incorrect scales, labels, etc.			S4

GCSE (9–1) content Ref.	Subject content	Initial learning for this qualification will enable learners to...	Foundation tier learners should also be able to...	Higher tier learners should additionally be able to...	DfE Ref.
12.03c	Bivariate data	Plot and interpret scatter diagrams for bivariate data. Recognise correlation.	Interpret correlation within the context of the variables, and appreciate the distinction between correlation and causation. Draw a line of best fit by eye, and use it to make predictions. Interpolate and extrapolate from data, and be aware of the limitations of these techniques.		S6
12.03d	Outliers	Identify an outlier in simple cases.	Appreciate there may be errors in data from values (outliers) that do not 'fit'. Recognise outliers on a scatter graph.		S4